LAB REPORT 02 - ADVERSARIAL ATTACKS

Introduction

In this lab, we explored adversarial attacks: perturbations applied to images that are imperceptible to the human eye yet cause a machine learning model to misclassify the input. The perturbation is crafted deliberately and mathematically, it is not random noise. Specifically, we implemented FGSM (Fast Gradient Sign Method), which exploits the linear nature of neural networks in high-dimensional spaces.

This is a relevant field of study today, particularly for privacy and protection from mass surveillance. I found myself getting interested in Fawkes, a tool developed by researchers at the University of Chicago, which adds imperceptible perturbations to personal photos before they are uploaded online, with the goal of making facial recognition systems fail to identify the person. A similar tool called Glaze applies the same principle to protect artists' styles from being scraped and learned by generative AI models.

White vs Black vs Grey attack and transferability

However, we cannot be fully certain that these tools work in all real-world scenarios. The protection is validated against known, public models, while the models deployed by surveillance companies are private and constantly being retrained. What we did in this lab is a **white-box attack**: we had full access to the model's architecture and weights, which allowed us to compute the exact gradient of the loss with respect to the input pixels. Real-world scenarios like Fawkes operate in a black-box setting instead, where the attacker can only observe the model's outputs. The reason researchers believe protection can still work in this setting is **transferability**: adversarial perturbations crafted against one model sometimes fool a different model too, particularly when the two share similar architecture or training data. It is not guaranteed, but it is frequent enough to be exploitable. This is the somewhat fragile assumption that tools like Fawkes are ultimately built on. Moreover, one established technique for making a model more robust is **adversarial training**: incorporating successful adversarial examples directly into the training data so the model learns to resist them. This creates an arms race: protection tools improve, models adapt, and the cycle continues.

How it works

The core idea of FGSM, introduced by Goodfellow, Shlens and Szegedy in 2014, is elegant: rather than changing pixels randomly, we ask the model itself which direction to push each pixel to maximise its own error. Concretely, we compute the gradient of the loss with respect to the input image, take only its sign (+1 or -1 per pixel), and add a small step of size ε in that direction. What makes the paper historically interesting is that Goodfellow reportedly wrote the code in a single night, and the idea emerged from a casual discussion. At the time, the dominant belief was that adversarial vulnerability came from the nonlinearity of neural networks. The paper argued the opposite: the vulnerability comes from their **linearity**, scaled into extremely high-dimensional space. In 784 dimensions (a 28×28 image), even a tiny per-pixel push accumulates into a large effect on the output, enough to flip the classification entirely. This reframes adversarial attacks not as a bug of complex models, but as an almost inevitable consequence of how linear operations scale.

What we did

The model we attacked was a LeNet-style CNN pretrained on MNIST, a dataset of 10 handwritten digits that we also used in the deep learning course. Before attacking, we ran a baseline accuracy check to confirm it was working correctly.

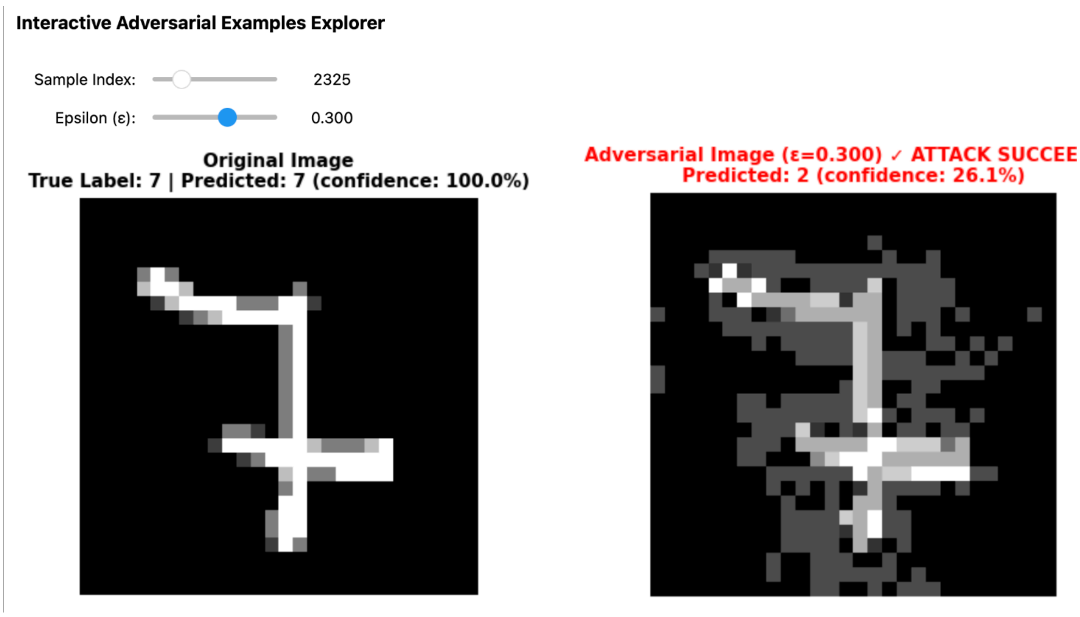
A line of code I found particularly interesting is `images.requires_grad = True`. Normally, gradients are computed with respect to the model's weights, but here we do something unusual: we compute the gradient with respect to the input pixels themselves, because what we want to modify is not the model but the image.

Interactive Adversarial Examples Explorer

The core tension of FGSM is finding an ε large enough to fool the model but small enough that a human would notice nothing unusual. Playing with the interactive explorer, I found several cases that illustrate this well.

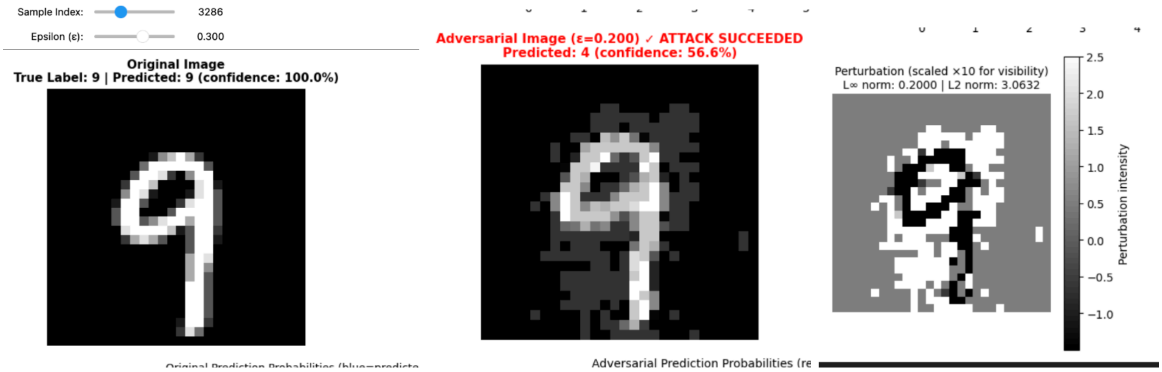
True label: digit 7. Predicted: digit 2

On sample 2325, the model predicted digit 2 instead of the true label 7 at ε = 0.3, with a confidence of 26.1%. The confusion is not arbitrary: a heavily perturbed 7 can start to resemble a 2 structurally when the horizontal stroke gets distorted. The low confidence also suggests the model was not strongly convinced: the attack pushed it over the decision boundary, but only just. At ε = 0.3 the perturbation is also starting to become visible to the human eye. At lower values, around 0.05 to 0.1, the two images are essentially indistinguishable, which is precisely the more concerning regime for real-world applications.



Digit 9 misclassified as digit 4

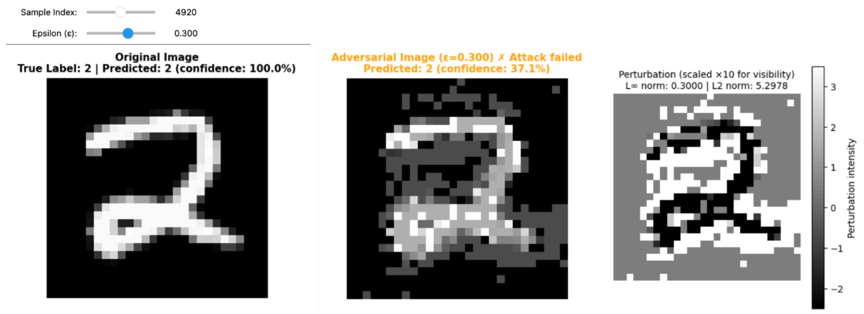
More interesting is sample 3286, a digit 9 misclassified as a 4 at the lower ε = 0.2, with 56.6% confidence. This is notable for two reasons. First, the attack succeeded at a smaller perturbation budget, suggesting this 9 sits closer to the decision boundary. Second, the 9/4 confusion has geometric logic: both digits share a closed upper region and a descending vertical stroke. The adversarial image still reads unmistakably as a 9 to the human eye, which illustrates the core property of FGSM: the perturbation crosses the model's decision boundary while remaining below the threshold of human perception. The perturbation map is also revealing, as the highest intensity pixels cluster around the loop and stroke of the 9, which are precisely the features the model finds most informative. FGSM is not adding noise randomly; it is targeting the structurally meaningful regions.



How robustness varies across digits

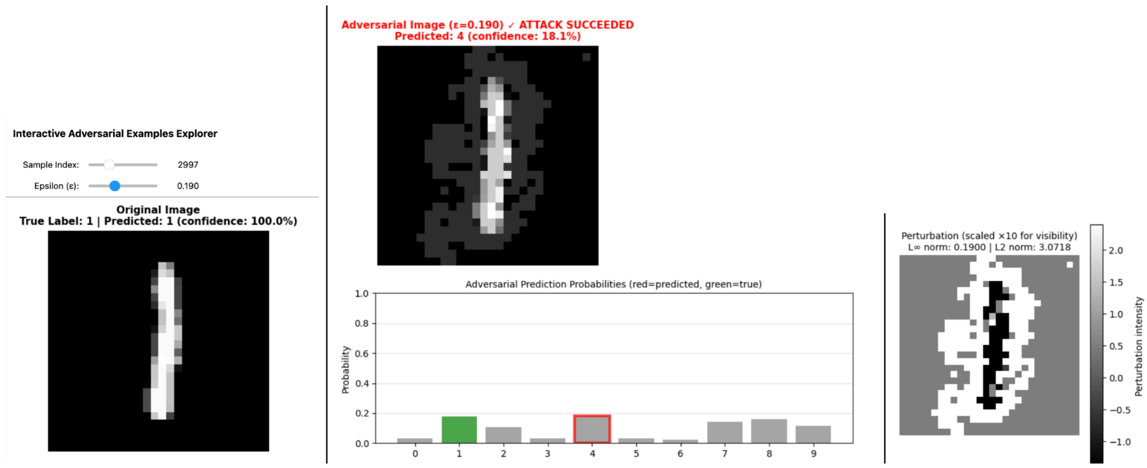
Digit 2 - distinctive shape - robust case

To better illustrate how robustness varies across digits, I searched for the most robust and most fragile examples. The digit 2 proved most robust: even at ε = 0.3 the attack failed entirely, with the model continuing to predict 2 at 37.1% confidence. The L2 norm of 5.2978 is the highest recorded across all experiments, meaning FGSM applied the largest total perturbation energy of any tested case and still could not cross the decision boundary. This likely reflects the geometric distinctiveness of the 2: its curved top, diagonal stroke and flat base make it sufficiently different from every other digit that even aggressive perturbation cannot push it convincingly toward another class.



Digit 1 - weak shape

The digit 1 tells the opposite story. Sample 2997 was misclassified as a 4 at ε = 0.19, the lowest successful attack epsilon across all experiments, with a confidence of just 18.1%. The probability distribution after the attack is almost flat, suggesting the model is not so much convinced it is a 4 as it is simply lost. A 1 is geometrically the simplest digit in MNIST: just a near-vertical stroke with very few active pixels. FGSM has little surface area to work with, but precisely because the digit is so minimal, even a small perturbation makes it ambiguous. A slightly altered vertical stroke could plausibly resemble a 1, a 4 or a 7, and the model has very little additional structure to anchor its prediction to.



Together these two cases illustrate that robustness under FGSM reflects two distinct factors: how geometrically distinctive the digit is, which determines how far it sits from neighboring decision boundaries, and how much visual information it contains, which determines how much the model has to hold onto when perturbation is applied.

These qualitative observations motivated a more systematic analysis. Rather than examining individual samples, the following section measures robustness across the entire test set, tracking how attack success rate, perturbation energy, and model confidence each evolve with ε. The patterns that emerged from the individual cases - the non-linearity of the tipping point, the plateau of unattackable samples, the concentration of perturbation in structurally meaningful pixels - all appear clearly in the aggregate metrics.